

TrES-3b

R-2635

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160930 · created 2026-08-29T00:48:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457661.6714 +/- 0.0013 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.194 +/- 0.018
Transit depth (Rp/Rs)^2	3.75 +/- 0.72 [%]
Orbital Inclination (inc)	82.27 +/- 1.22
Airmass coefficient 1 (a1)	1.027 +/- 0.015
Airmass coefficient 2 (a2)	-0.016 +/- 0.0031
Scatter in the residuals of the lightcurve fit is	1.45 %
Best Comparison Star	#1 - [282, 333]
Optimal Aperture	3.66
Optimal Annulus	10.39
Transit Duration (day)	0.061 +/- 0.011

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.194 ± 0.018 vs 0.16309 (deviation 1.51σ)
Duration fit vs published	0.061 d vs 0.0567 d (deviation 0.35σ)
Mid-time O–C	-2.8 min
Depth SNR	9.4
Residual scatter	1.45 %

Light curve

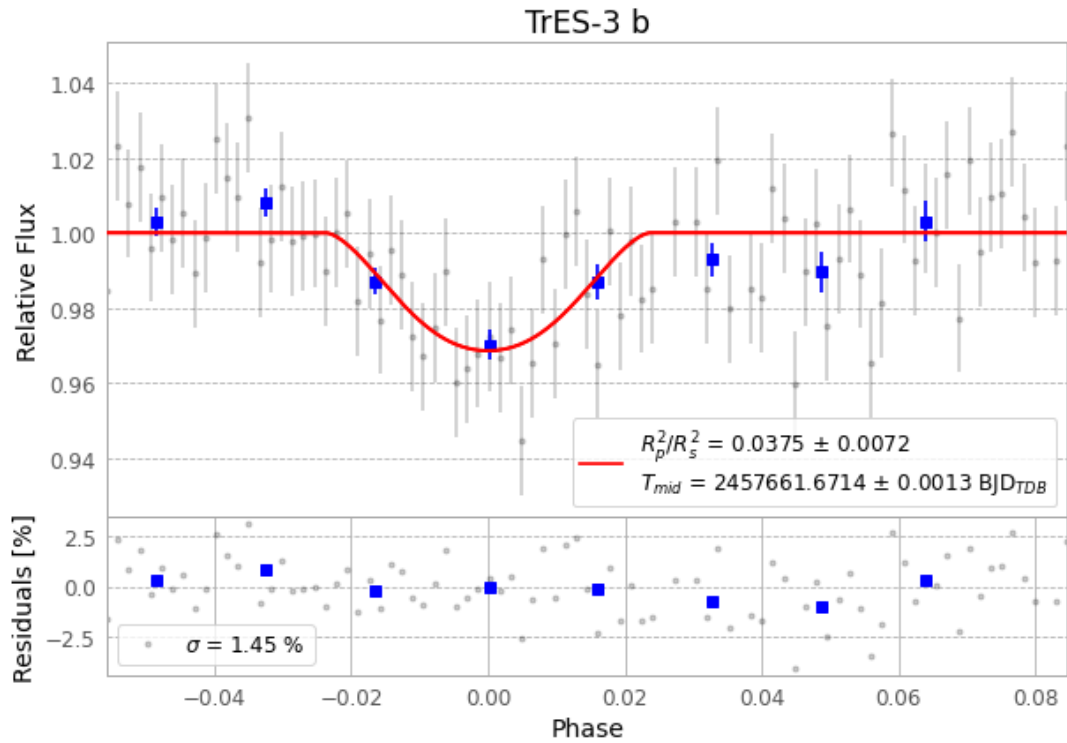


Figure 1 — FinalLightCurve_TrES-3 b_2016-09-29.png (SHA-256 5095ca49bc7bbccd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [351, 248], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 b4dc6b28f358d743...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

87 FITS frames (57 MB), TRES-3160930022112.FITS → TRES-3160930064512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160930.log (39d9ed1b2015...), FinalLightCurve_TrES-3 b_2016-09-29.png (5095ca49bc7b...), FinalLightCurve_TrES-3 b_2016-09-29.csv (283ae2bbfb4f...)

Compute resources

Wall time 117.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)